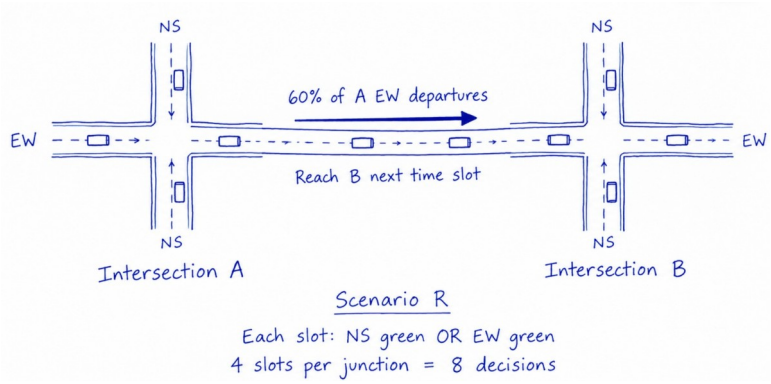
Quantum Traffic Gridlock

We started with a base uniform QUBO paired with standard QAOA model. We improved its quantum circuit, then improved the traffic model itself. We selected a weighted QUBO with gradient trained QAOA. It performed better across varied tests.

The problem:

- We control two linked intersections, A and B, over four time slots.
- In each slot, we choose north–south or east–west green at each junction.
- This gives eight binary decisions and 256 possible schedules.
- Traffic leaving A can reach B in the next slot. Queues also carry over between slots. Our decisions must account for both effects.



We aim to reduce the official traffic cost. It combines:

- Delay from queues left after service in each slot.
- Overflow penalties when queues exceed storage.
- Switching penalties when the green direction changes between slots.

How we used QUBO and QAOA

We used a QUBO to approximate traffic cost. We then used QAOA to sample schedules with low model cost.

- A 1 means north–south green. A 0 means east–west green. The first four bits belong to A; the last four belong to B.
- The QUBO coefficients were derived from queue dynamics. They assign costs to single decisions and pairs of decisions. We first used a uniform fit, then gave low-cost schedules more weight.
- We mapped the QUBO to a quantum cost operator, using one qubit per decision. All schedules are allowed, so we needed no extra validity penalties.
- Then did the circuit with equal weight on all schedules. We alternated cost operations with X mixer rotations. A classical optimizer trained the angles to lower expected QUBO cost.

Our QUBO is an approximation, so low model cost does not guarantee the lowest traffic cost.

How we benchmark against optimum values

We find the exact optimum by checking every possible traffic schedule. Our problem has eight binary decisions, with each decision choosing N-S or E-W green at an intersection during a time slot. This gives us 256 possible schedules. We generate all of them and calculate their traffic costs using the official scorer, which accounts for queue delay, overflow and signal switching. The lowest cost across these schedules is the true optimum. We found this to be 72 for scenario R and 234 for scenario H.

Then we use these optimum values to benchmark our QAOA results. We train the circuit to reduce its expected QUBO cost and measure it 4,096 times for each of five fixed seeds. Every measurement produces a schedule, which we evaluate using the official scorer. We include all measurements when calculating the average cost, rather than reporting only the best schedule found. We compare this average against the exact optimum and calculate lift, which measures how much of the gap between random scheduling and optimal scheduling our solver closes. QAOA does not receive the optimal schedule or optimum value during training. However, our weighted QUBO construction already evaluates all 256 traffic histories before fitting, so the classical preprocessing contains enough information to find the optimum itself.

About Version1

We built the first model with a uniform quadratic projection (degree-two Walsh projection). Here all schedules have equal importance when approximating their traffic costs.

- We computed the coefficients from conditional expectations of queue dynamics.
- We used standard QAOA with eight qubits, three layers and six angles.
- We trained the angles with COBYLA. This optimizer tries changes without using gradients.
- On public R scenario, it gave mean cost 89.893. Uniform random schedules gave 151.25. The true optimum was 72.

We improved the circuit and its training

- Increased the depth to four layers. R cost fell to 88.717
- Added analytic gradients with a shared mixer angle per layer. R cost fell to 86.581.
- Then gave each decision its own X mixer angle in each layer. This let the circuit adjust different junctions and slots separately.

Our improved circuit had four cost angles and 32 mixer angles. We calculated gradients using adjoint differentiation. We trained with L-BFGS-B (similar to Adam W optimizer in ML), within the same 300-evaluation budget.

- R cost fell to 75.282. Optimal samples rose from 6.37% to 67.08%.

Our results with the weighted model

We used the official scorer. Each mean includes all 4,096 raw measurements per seed, averaged across five seeds, 0–4. We did not filter or repair samples.

Scenario	True optimum	Uniform QUBO	Weighted QUBO
Scenario R	72.0	75.282	75.484
Scenario H	234.0	236.477	235.749

Both columns use the improved QAOA. This isolates the QUBO change. We accepted a small R regression because we saw wider gains.

We also tested unseen H.

  - Our mean cost was 235.749, only 0.75% above the optimum of 234. Greedy control gave 244. The best alternating schedule gave 261.
- Mean lift was 97.36%. Exactly optimal measurements made up 29.18% of our samples. Every seed sampled the optimum.

Conclusion

We improved the search first, then the model it searched. Independent mixer angles and gradients helped us use the original QUBO better. Boltzmann weighting then improved accuracy around useful schedules.

Our method is strong within the tested two-intersection, four-slot format. It does not win everywhere.

We verified saved OpenQASM 3 circuits, reproduced seeded counts and recomputed metrics. Our QUBO study verified 915 circuit runs. The H comparison added 10.